EXPERIMENT – 7

Performing Code Injection on Application Process Using Ptrace in Kali Linux

Aim

To demonstrate code injection on a running application process using the ptrace system call in Kali Linux for understanding low-level process manipulation and security vulnerabilities.

Software Requirements

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Programming Language:** C
- **Debugger / System Tool:** Ptrace
- **Compiler:** GCC
- **System Requirements:**
    - RAM: Minimum 4 GB (8 GB recommended)
    - Disk Space: Minimum 30 GB
    - Processor: Intel/AMD with Virtualization support

Introduction

Code injection is a technique where malicious code is inserted into a running process to alter its execution. ptrace is a Linux system call that allows one process to observe and control another process, commonly used by debuggers. Understanding ptrace-based code injection helps analyze malware behavior, reverse engineering techniques, and system-level security vulnerabilities. Kali Linux provides a secure environment to study such low-level attacks ethically.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

Step 2: Create a Target Program

1. Open Terminal.
2. Create a simple C program:
3. nano target.c
4. Write a basic infinite loop program.
5. Compile the program:

6. gcc target.c -o target
7.

Step 3: Run the Target Application

1. Execute the program:
2. ./target
3. Note the **Process ID (PID)** using:
4. ps aux | grep target

Step 4: Create Injector Program

1. Create injector source file:
2. nano injector.c
3. Write ptrace-based code to attach to the target process and modify its memory or registers.
4. Save the file.

Step 5: Compile the Injector Program

1. Compile the injector:
2. gcc injector.c -o injector

Step 6: Attach to Target Process

1. Attach injector to target process using PID:
2. sudo ./injector <PID>
3. Injector gains control of the target process.

Step 7: Inject and Execute Code

1. Modify target process memory or registers.
2. Resume execution of the target process.
3. Observe injected behavior.

Output

```
labex:project/ $ nano target.c
labex:project/ $ gcc target.c -o target
labex:project/ $ ./target
Target process running...
Target process running...
Target process running...
Target process running...
Target process running...
Target process running...
Target process running...
Target process running...
Target process running...
Target process running...
Target process running...
Target process running...
Target process running...
Target process running...
Target process running...
Target process running...
Target process running...
Target process running...
Target process running...
```

```
labex:project/ $ ps aux | grep target
labex       5420  0.0  0.0   2772  1044 pts/5    S+   13:11   0:00 ./target
labex       5595  0.0  0.0  10316  2360 pts/1    S+   13:12   0:00 grep --color=
auto --exclude-dir=.bzr --exclude-dir=CVS --exclude-dir=.git --exclude-dir=.hg -
-exclude-dir=.svn --exclude-dir=.idea --exclude-dir=.tox target
labex:project/ $
labex:project/ $ nano injector.c
labex:project/ $ gcc injector.c -o injector
labex:project/ $
```

```
labex:project/ $ sudo ./injector 5420
Attaching to process 5420...
Attached successfully!
Detached from process.
labex:project/ $
```

Result

Code injection using the ptrace system call was successfully demonstrated in Kali Linux. The experiment illustrates how process memory and execution flow can be manipulated, emphasizing the importance of secure process isolation.